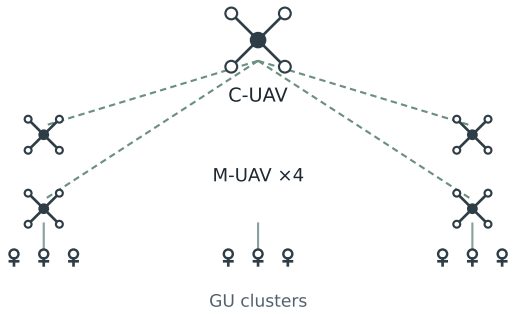
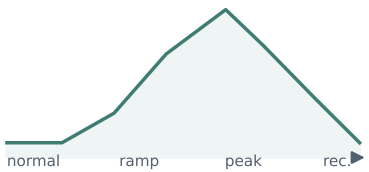
A. Environment Module

A1 System model

500 m × 500 m, K = 4; 5/20 MHz access/backhaul



A2 QoS session model



20 baseline
→ up to 130 users

QoS tiers
0.7 / 2.0 / 2.5 Mbps

session 45–120 s

A3 UAV / GU dynamics

GU: 0–6 m/s, resampled every 0.5 s

M-UAV: K-means++ + Hungarian,
25 m/s, 5 m/s^2

Age: $O_i \leftarrow O_i + \Delta t$ if QoS missed,
else $O_i \leftarrow 0$



position → association → age

B. Branch-Credit SAC

s_t

B1 Shared relay set encoder

relay tokens ×4

permutation-invariant encoder

B2 Motion branch

$\pi_M(s) \rightarrow a_M$

$Q_M(s, a_M)$ recursive long-horizon

a_M

B3 Power branch

$\pi_P(s) \rightarrow a_P$

$Q_P(s, a_M, a_P)$ immediate same-slot

B4 SAC update

replay buffers
(main + power)
target networks
$L_Q,\ L_\pi,\ L_\alpha$

a_M

a_P

state s_t: C-UAV + relay tokens + QoS/age/load

long arrow = recursive credit; short arrow = immediate credit

r_t

r_t

C. ROCS-0 Scheduler

C1 Age-weighted priority

$w_i = 1 + \lambda_s(O_i + \Delta t)$

$A_i = q_i / C_i^A, \quad B_i = q_i / R_k^B$

density $= w_i / \max(A_i, B_i)$

λ_s: resilience knob

C2 Max-cost + 1-swap

1) greedy feasible pass

2) 1-swap improvement

3) refill by density

access + backhaul feasible

C3 Allocate & update

served bits z_i → per-relay power

per-relay backhaul, QSR

update age O_i; next slot

$\sum_i$ access ≤ 1, $\sum_i$ backhaul $\le R_k^B$

z_t (allocation) → environment